Chaehyeong Lee

Boulder, Colorado, USA

+1 (303) 258-6841 • Chaehyeong.Lee@colorado.edu

sites.google.com/yonsei.ac.kr/hyeong • https://orcid.org/0009-0005-3110-9839

Research Interests

Geofluid Dynamics: Cross-scale upper-ocean mixing processes; Mesoscale eddies.

Physical Oceanography: Long-term change in the upper-ocean thermal state; Ocean's role in climate systems.

Education

University of Colorado Boulder — **Boulder, CO, USA**

Ph.D. in Atmospheric and Oceanic Sciences — *Aug. 2024 – present*

Advisor: Dr. Donata Giglio & Co-Advisor: Dr. Aneesh Subramanian

Yonsei University — **Seoul, Rep. of Korea**

M.S. in Atmospheric Sciences — *Aug. 2023*

Advisor: Dr. Hajoon Song

Yonsei University — **Seoul, Rep. of Korea**

B.S. in Atmospheric Sciences — *Feb. 2022*

Advisor: Dr. Hajoon Song

Publications

Under Review

1. **Lee, C.**, Giglio, D., Subramanian, A. C., Han, W., Capotondi, A., Du, D., Molod, A., & Ruiz-Xomochuk, V. (2026). Effects of Satellite Sea Surface Salinity Assimilation on the Representation of Upper-Ocean Thermohaline Structure and Variability in the NASA GEOS-S2S-2 model. *Quarterly Journal of the Royal Meteorological Society*

In Preparation

1. **Lee, C.**, Giglio, D., Subramanian, A. C., & Arbic, B. K.

Published

1. **Lee, C.**, Song, H., Choi, Y., Cho, A., & Marshall, J. (2025). Observed Multi-decadal Increase in the Surface Ocean's Thermal Inertia. *Nature Climate Change*, **15**, 308-314. doi:10.1038/s41558-025-02245-w

Research Experience

Dr. Giglio's Research Group, CU Boulder — **Boulder, CO, USA**

Research Assistant — *Aug. 2024 – present*

Climate Processes and Predictability Group, CU Boulder — **Boulder, CO, USA**

Research Assistant — *Aug. 2024 – present*

Air-Sea Modeling Lab, Yonsei University — **Seoul, Rep. of Korea**

Research Assistant — *Dec. 2020 – Aug. 2024*

Teaching Experience

Data Science Laboratory — **Boulder, CO, USA**
Teaching Assistant — *Spring 2026*

Climate & Civilization — **Seoul, Rep. of Korea**
Teaching Assistant — *Spring 2023*

Physical Oceanography — **Seoul, Rep. of Korea**
Teaching Assistant — *Fall 2022*

Awards & Scholarships

2025: ATOC Fellowship, Department of Atmospheric and Oceanic Sciences, University of Colorado Boulder

2025: Academic Research Grants (GCP research credits), Google LLC

2024: Outstanding Thesis Award, Yonsei University Graduate School, Yonsei University

2022–2023: Full tuition merit scholarship (for the top 2 graduate students), Yonsei University

2022: High Honors for academic performance, Yonsei University

2020–2021: Jilli Scholarship for academic performance, Yonsei University

Patent

Song, H., & **Lee, C.** (2025). *Evaluation system and method of persistence of SST anomalies using autocorrelation coefficient and arctangent regressive model*. Rep. of Korea Patent #1028135790000. doi:10.8080/1020220157159

Invited Talk

NASA Salinity Telecon — **Virtual Meeting**
Assessing the Impact of Satellite Sea Surface Salinity Assimilation on Vertical Structure of the Upper Ocean in the NASA GEOS-S2S 2. — *Dec. 2025*

Selected Presentations

2027 American Meteorological Society Annual Meeting (poster) — **Denver, CO, USA**
Ocean Advection Drives Mixed Layer Temperature Variance at 10-100 km in the Gulf Stream Extension. — *Jan. 2027 (scheduled)*
***Lee, C.**, Giglio, D., & Subramanian, A. C.*

US Climate Variability and Predictability (poster) — **Boulder, CO, USA**
Enhanced Upper-Ocean Horizontal Advection in a High-Resolution Climate Model Improves the Representation of High-Frequency Near-Surface Temperature Variability. — *Sep. 2026*
***Lee, C.**, Giglio, D., & Subramanian, A. C.*

Ocean Science Meeting 2026 (poster) — **Glasgow, Scotland**
Assessing the Impact of Satellite Sea Surface Salinity Assimilation on the Upper Ocean Thermal State in the NASA GEOS S2S-v2 Model. — *Feb. 2026*
***Lee, C.**, Giglio, D., Subramanian, A. C., Han, W., Capotondi, A., Du, D., & Molod, A.*

American Geophysical Union Fall Meeting (poster) **Chicago, IL, USA**
The Increasing Trend of Persistence of Sea Surface Temperature in the Past 40 Years. *Dec. 2022*
***Lee, C.**, Song, H., Cho, A., & Tak, Y.*

Korean Society of Oceanography Spring Conference (talk) **Jeju, Rep. of Korea**
Increasing Persistence of SST Anomalies and Duration of Marine Heatwaves. *Jun. 2022*
***Lee, C.**, Song, H., Cho, A., & Tak, Y.*

Workshops

Broadening Earth System Research Across Scales with High-Resolution Modeling Workshop **Boulder, CO, USA**
US Climate Variability and Predictability *Sep. 2026*

Data Assimilation Training and Science Workshop **Norman, OK, USA**
Consortium for Advanced Data Assimilation Research and Education & NOAA EPIC *Jun. 2026*

14th International Workshop on Modeling the Ocean 2024 **Sapporo, Japan**
International Workshop on Modeling the Ocean *Jun. 2024*

User Training for the Glosea 6 Climate Prediction Model **Jeju, Rep. of Korea**
Korea Meteorological Administration *Jan. 2022*

Deep Learning Training: Fundamentals of Deep Learning **Gonju, Rep. of Korea**
NVIDIA Deep Learning Institute *Jan. 2022*

Academic Service

Peer Reviewer: *Journal of Climate*

Technical Skills

Programming: Python (xarray, dask, Pangeo), Julia
Modeling: MITgcm, Oceananigans
HPC: Parallel/distributed computing, NCAR Casper/Derecho clusters
Tools: Git, Linux shell scripting, LaTeX